Autocorrelation of a Rectangularly Windowed Linear Chirp

Hein Hundal*

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Let $B > 0$ be the swept bandwidth and let $T_p > 0$ be the pulse duration. Using $j^2 = -1$, define the chirp

$$f(t) = \exp\left[j\left(\frac{\pi B}{T_p}t^2 - \pi B t\right)\right](u(t) - u(t - T_p)), \tag{1}$$

where $u(t)$ is the unit-step function. Thus $f(t)$ is supported on $[0, T_p]$. The values assigned to the step functions at the endpoints do not affect any integral below.

We use the autocorrelation convention

$$R_f(\tau) = \int_{-\infty}^{\infty} f(t) f^*(t - \tau)\, dt \tag{2}$$

and the normalized sinc function

$$\operatorname{sinc}(x) = \begin{cases} \dfrac{\sin(\pi x)}{\pi x}, & x \neq 0, \\ 1, & x = 0. \end{cases} \tag{3}$$

The overlap interval

The first factor in the integrand is nonzero when $0 \le t \le T_p$. The second is nonzero when

$$\begin{aligned} 0 &\le t - \tau \le T_p, \\ \tau &\le t \le T_p + \tau. \end{aligned} \tag{4}$$

The product must be zero if

$$\min(T_p, T_p + \tau) \le t \quad \text{or} \quad t \le \max(0, \tau).$$

If $|\tau| \ge T_p$, then

$$\begin{aligned} \min(T_p, T_p + \tau) \le \max(0, \tau) \quad &\text{so,} \\ R_f(\tau) = 0 \quad &\text{if } |\tau| \ge T_p. \end{aligned} \tag{5}$$

Futhermore, the two factors overlap on

$$I_\tau = [\max(0, \tau),\ \min(T_p, T_p + \tau)]. \tag{6}$$

Now suppose $|\tau| \le T_p$. The length and midpoint of the overlap interval are, respectively,

$$\begin{aligned} L(\tau) &= \min(T_p, T_p + \tau) - \max(0, \tau) = T_p - |\tau|, \\ c(\tau) &= \frac{\max(0, \tau) + \min(T_p, T_p + \tau)}{2} = \frac{T_p + \tau}{2}. \end{aligned} \tag{7}$$

Thus $I_\tau = [c(\tau) - L(\tau)/2, c(\tau) + L(\tau)/2]$.

*I did the math. GPT wrote most of the paper based on my math.

Evaluation of the autocorrelation

Write the phase of the unwindowed chirp as

$$\phi(t) = \frac{\pi B}{T_p} t^2 - \pi B t. \tag{8}$$

On the overlap interval, the phase in the product $f(t)f^*(t-\tau)$ satisfies

$$\begin{aligned} \phi(t) - \phi(t-\tau) &= \frac{\pi B}{T_p}\left[t^2 - (t-\tau)^2\right] - \pi B\left[t - (t-\tau)\right] \\ &= \frac{\pi B}{T_p}(2t\tau - \tau^2) - \pi B\tau \\ &= \frac{2\pi B\tau}{T_p}\left(t - \frac{T_p + \tau}{2}\right) \\ &= \frac{2\pi B\tau}{T_p}(t - c(\tau)). \end{aligned} \tag{9}$$

Make the centered change of variable $x = t - c(\tau)$. By (8), its limits are $-L(\tau)/2$ and $L(\tau)/2$. For $0 < |\tau| \le T_p$, the autocorrelation is therefore

$$\begin{aligned} R_f(\tau) &= \int_{I_\tau} \exp\big(j[\phi(t) - \phi(t-\tau)]\big)\, dt \\ &= \int_{-L(\tau)/2}^{L(\tau)/2} \exp\left(j\frac{2\pi B\tau}{T_p}x\right) dx \\ &= \frac{T_p}{j2\pi B\tau} \exp\left(j\frac{2\pi B\tau}{T_p}x\right)\Bigg|_{-L(\tau)/2}^{L(\tau)/2} \\ &= \frac{T_p}{\pi B\tau} \sin\left(\frac{\pi B\tau L(\tau)}{T_p}\right) \\ &= L(\tau)\ \mathrm{sinc}\left(\frac{B\tau L(\tau)}{T_p}\right). \end{aligned} \tag{10}$$

At zero lag, direct evaluation gives

$$\begin{aligned} R_f(0) &= \int_{-\infty}^{\infty} |f(t)|^2\, dt \\ &= \int_0^{T_p} 1\, dt \\ &= T_p. \end{aligned} \tag{11}$$

This is also the continuous limit of (11), since $L(0) = T_p$ and $\mathrm{sinc}(0) = 1$.

Combining the overlap and non-overlap cases gives the exact result

$$\boxed{R_f(\tau) = \begin{cases} (T_p - |\tau|)\ \mathrm{sinc}\left(\dfrac{B\tau(T_p - |\tau|)}{T_p}\right), & |\tau| \le T_p, \\ 0, & |\tau| > T_p. \end{cases}} \tag{12}$$

Equivalently, for $0 < |\tau| \le T_p$,

$$\begin{aligned} R_f(\tau) &= \frac{T_p}{\pi B \tau} \sin\left[\pi B \tau \left(1 - \frac{|\tau|}{T_p}\right)\right] \\ &= \frac{T_p}{\pi B |\tau|} \sin\left[\pi B |\tau| \left(1 - \frac{|\tau|}{T_p}\right)\right]. \end{aligned} \tag{13}$$

The second equality follows because the sine function is odd.

The sinc argument

$$\frac{B\tau(T_p - |\tau|)}{T_p} = B\tau\left(1 - \frac{|\tau|}{T_p}\right) \tag{14}$$

is piecewise quadratic in τ. Thus the result is a triangular overlap factor, $T_p - |\tau|$, multiplied by a sinc having a piecewise-quadratic argument. In particular, it is not merely a sinc by itself. The result is real and even, as is also clear from (13).

For a large time-bandwidth product BT_p, the first positive null near the origin is close to $1/B$. More precisely, the central null solves

$$B\tau\left(1 - \frac{\tau}{T_p}\right) = 1, \tag{15}$$

so, when $BT_p \ge 4$, it is

$$\tau_{\text{null}} = \frac{T_p}{2}\left(1 - \sqrt{1 - \frac{4}{BT_p}}\right) \approx \frac{1}{B}. \tag{16}$$